Cardiac Risk Visualizer: Technical Report

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Multimodal AI Hackathon 2026, Track A: Cardiovascular Risk Visualization & Prediction

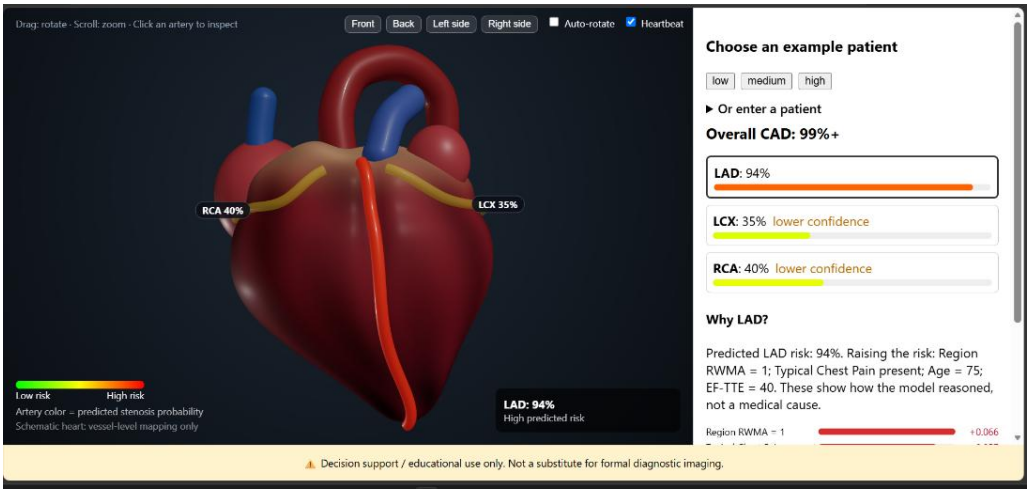
Code: https://github.com/princy21aj/cardio-risk-visualizer

> *Decision support and educational use only. Not a substitute for formal diagnostic imaging.*

1. Summary

Cardiac Risk Visualizer predicts overall coronary artery disease (CAD) and the stenosis status of the three main coronary arteries (LAD, LCX and RCA) from routine clinical data, and shows the result on an interactive 3D heart. For each prediction the dashboard explains which patient factors raised or lowered the risk (SHAP values) and states how far the prediction can be trusted. The CAD model reaches a cross-validated ROC-AUC of 0.92. The LCX and RCA models are weaker (AUC about 0.72), and the application says so openly.

Figure 1. The application: example patients, 3D heart, vessel probabilities and explanation panel.



2. Dataset and preprocessing

Data. We used the UCI "Extension of Z-Alizadeh Sani Dataset" (303 patients, CC BY 4.0). It contains demographic, examination, laboratory, ECG and echocardiographic features, plus the overall CAD label and the LAD, LCX and RCA stenosis labels. In this dataset CAD means at least 50% narrowing of one or more major coronary vessels.

Preprocessing. The cleaned table (data/clean.csv) has 303 rows and 60 columns: 56 input features and 4 targets (CAD, LAD, LCX, RCA). It has no missing values and all columns are numeric. Yes/no findings are stored as 0/1 indicators, bundle branch block is one-hot encoded (BBB_LBBB, BBB_N, BBB_RBBB), and ordinal codes such as Region RWMA and Function Class are kept as numbers. Measurements such as age, blood pressure, cholesterol, EF-TTE and blood counts keep their original units. Random Forests do

not need feature scaling, so values were not scaled, which also lets the dashboard show real values such as "Age = 75". The share of patients with a positive label is 71% for CAD, 58% for LAD, 39% for LCX and 38% for RCA.

Leakage prevention. The columns LAD, LCX, RCA and Cath are removed from every model's inputs, as the dataset authors and the hackathon brief require. They are the labels being predicted, or come directly from the angiography that produced them. All four models share the same 56-feature input table, and the backend lists the inputs through its /features endpoint.

3. Model architecture and training

We trained four separate Random Forest classifiers, one per target.

Target	Settings
CAD, LAD	300 trees, random seed 42
LCX, RCA	300 trees, minimum 3 samples per leaf, balanced class weights, random seed 42

LCX and RCA are positive in only about 38 to 39% of patients, so we used balanced class weights for them. The overall CAD model is trained directly on the CAD label and is not computed from the vessel models. Because the models are separate, the CAD probability can occasionally be lower than a vessel probability.

Validation. Metrics come from repeated stratified 5-fold cross-validation with 3 repeats (15 fits per target). The final models are then fitted on all 303 patients for the application. There is no separate held-out test set, so the reported scores are cross-validated estimates on one dataset. The script backend/train.py reproduces the models, the demo patients and metrics.json in one run.

4. 3D visualisation pipeline

The frontend uses React, Vite, Three.js and react-three-fiber, and runs in a normal browser (tested on a standard laptop in Edge and Chrome).

- **Heart.** A sphere mesh is reshaped by a mathematical function into a heart: tapered rounded apex, larger left ventricle, flatter back, and grooves between the ventricles and between atria and ventricles. Vertex colours give deeper muscle red at the bottom and lighter tissue near the base. No external mesh is needed.
- **Anatomy.** Right atrium, left atrial appendage, aorta, pulmonary artery and vena cava are placed around the heart.
- **Coronary arteries.** LAD, LCX and RCA are tubes that follow paths on the heart surface (LAD in the front groove, LCX and RCA along the atrioventricular groove).
- **Risk mapping.** Each artery's colour is a gradient from green (low) through yellow and orange to red (high), set from that artery's predicted stenosis probability. Arteries with 60% or more, and the selected artery, pulse softly.

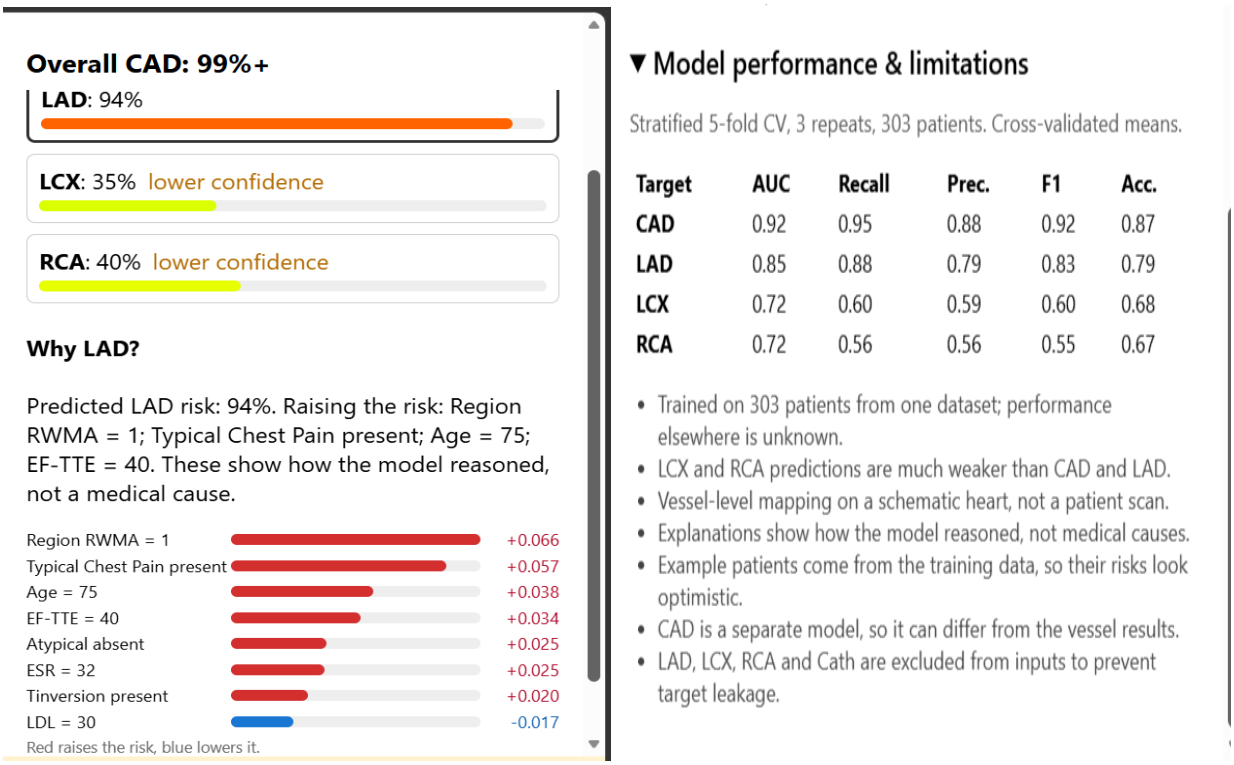
- **Interaction.** Users rotate, zoom, use camera presets (front, back, left, right), toggle auto-rotate and heartbeat, and click an artery to inspect it. The selected artery is highlighted and the others dim.

5. Clinical dashboard and explanations

The right-hand panel shows:

- **Patient input:** three example patients (low, medium and high risk) or a form for entering values. Missing values are filled with the dataset median, and the response lists which values were assumed.
- **Probabilities:** overall CAD, LAD, LCX and RCA, with a "lower confidence" tag on vessels whose cross-validated AUC is below 0.75 (LCX and RCA).
- **Explanation:** the eight features with the largest SHAP contribution for the selected artery, drawn as red bars (raise the risk) and blue bars (lower the risk) next to the patient's actual value, plus a plain-language sentence. The text states that SHAP shows how the model reasoned, not a medical cause.
- **Model performance and limitations:** the cross-validated metrics table and a list of limitations.
- **Safety disclaimer:** a visible banner at the bottom of the screen.

Figure 2. SHAP explanation and the built-in performance panel.



6. System architecture and usage

The React frontend calls a FastAPI backend (Uvicorn). The backend loads the saved models and a SHAP TreeExplainer for each target. Endpoints: /features (inputs and defaults), /demo-patients, /predict (probabilities, explanations and assumed defaults for one patient) and /metrics (the table shown in the dashboard). New features, models or anatomical structures can be added without redesigning the application, because the models, the API response and the 3D arteries are separate parts.

Run it

1. Backend: create a virtual environment, run `pip install -r requirements.txt`, then `uvicorn main:app --reload` in the backend folder (http://127.0.0.1:8000).
2. Frontend: in the frontend folder run `npm install` and `npm run dev` (http://localhost:5173).
3. Retrain (optional): from the project root run `python backend/train.py`.

7. Evaluation results

Table 1. Cross-validated performance (mean over 15 folds, with standard deviation)

Target	Accuracy	Precision	Recall	F1	ROC-AUC
CAD	0.873 (0.041)	0.882 (0.037)	0.952 (0.034)	0.915 (0.027)	0.923 (0.031)
LAD	0.786 (0.042)	0.785 (0.046)	0.880 (0.061)	0.827 (0.034)	0.851 (0.045)
LCX	0.680 (0.040)	0.594 (0.053)	0.605 (0.059)	0.597 (0.044)	0.725 (0.068)
RCA	0.667 (0.046)	0.561 (0.075)	0.555 (0.118)	0.552 (0.076)	0.718 (0.052)

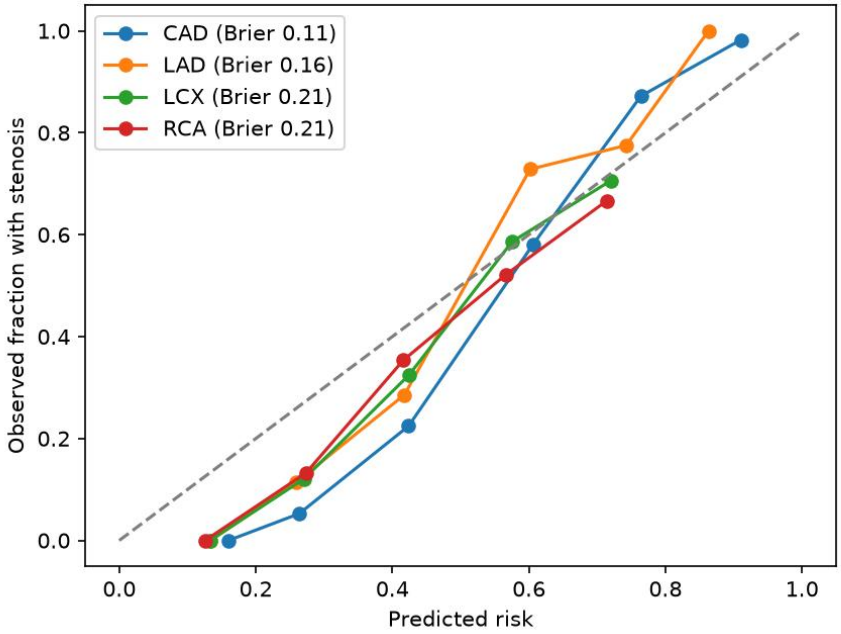
To judge accuracy fairly, compare it with always predicting the most common class, which would score about 0.71 for CAD, 0.58 for LAD, 0.61 for LCX and 0.62 for RCA. CAD (0.873) and LAD (0.786) clearly beat this baseline, while LCX (0.680) and RCA (0.667) are only 5 to 7 points above it. The CAD model finds about 95 of every 100 true CAD cases and LAD about 88. LCX and RCA find only about 55 to 60 of every 100 true cases, so their outputs should be read as rough.

Table 2. Calibration (Brier score, lower is better)

Target	Brier score
CAD	0.11
LAD	0.16
LCX	0.21
RCA	0.21

A Brier score of 0.25 equals always predicting 50%, so LCX and RCA are only slightly better than guessing. The CAD curve lies below the diagonal in the middle range (about 42% predicted against about 24% observed), so CAD overestimates risk there. We did not apply isotonic recalibration because 303 patients are too few to fit it reliably.

Figure 3. Calibration curves for the four targets, 5-fold cross-validation.



8. Limitations

- Only 303 patients from one dataset, with no external validation.
- LCX and RCA are much weaker than CAD and LAD.
- CAD overestimates risk in the middle probability range.
- CAD is a separate model, so it can occasionally be lower than a vessel prediction.
- The heart is a schematic. The mapping is vessel-level, not a patient scan or lesion localisation.
- SHAP values describe the model's reasoning, not medical causes.
- The example patients are taken from the training data, so their risks look optimistic (for example the high-risk patient shows an overall CAD above 99%).
- The application is for decision support and education only.

9. Conclusion and future work

We built an end-to-end, leakage-safe pipeline from clinical data to an explainable 3D risk display, with honest reporting of performance, calibration and limits. Future work: validate on an external cohort, improve LCX and RCA with more data and other models, recalibrate when more data is available, and link predictions to real imaging.

References

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2. Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems 30.

3. Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32.
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